PERINATAL INFECTIONS

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INTRODUCTION

- Important cause of still births and morbidity
- Many diseases go undiagnosed
- Appropriate treatment can prevent morbidity/mortality
- 1971: Andres Nahmias proposed acronym ToRCH
- 1975: Harold Fuerst added Syphilis to the acronym

ACRONYM:**TORCHES CLAP**

Toxoplasmosis
Other
(Syphilis,Varicella Zoster,
ParvovirusB- 19, HEP B)

Rubella
CMV
Herpes simplex
Enterovirus
Syphilis

- Latest addition: Zika virus

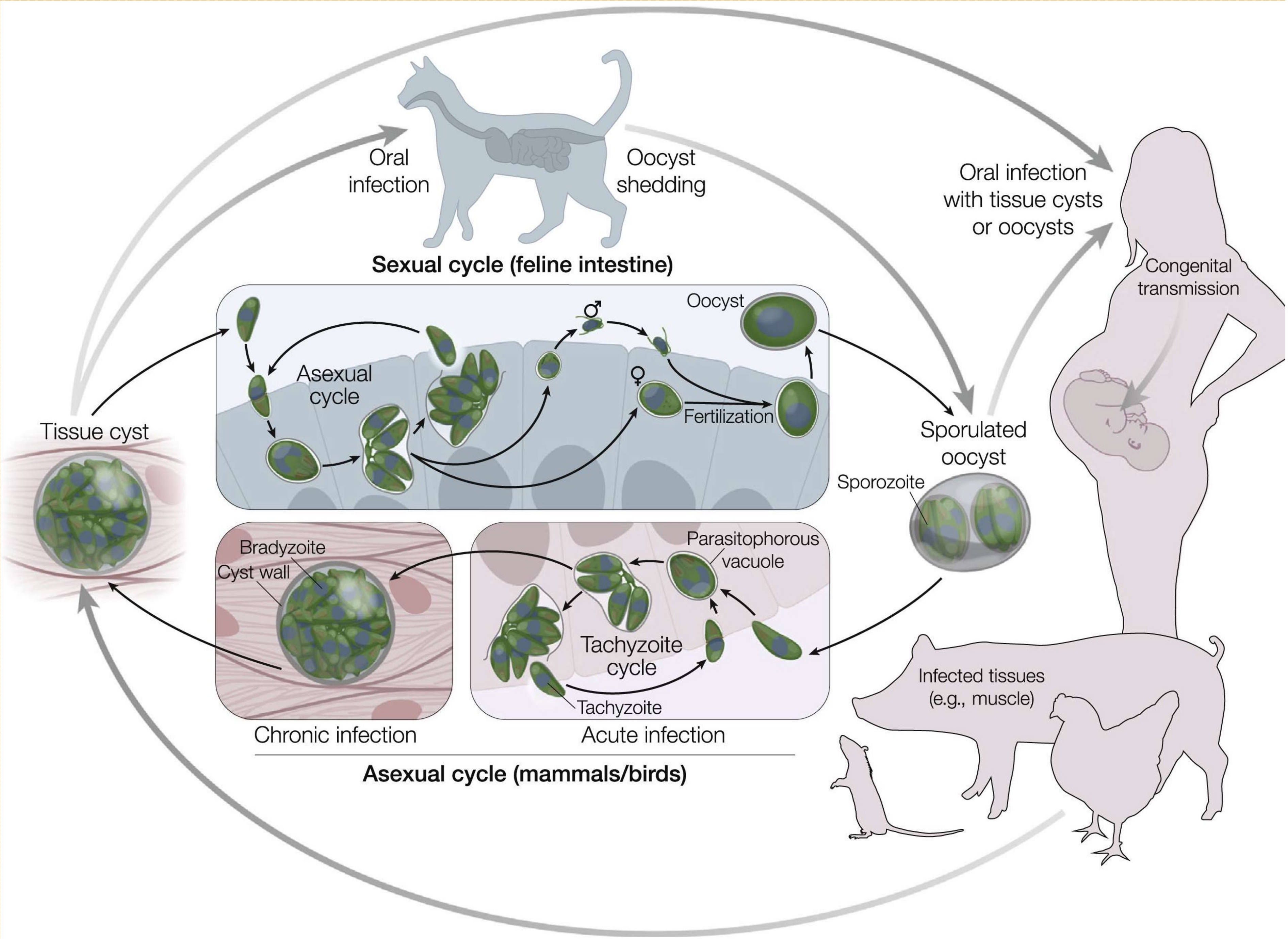
Chickenpox
Lyme disease
AIDS
Parvovirus B19

Toxoplasmosis

- A disease caused by an intracellular parasite *Toxoplasma Gondii*.

- Human infection occurred by: Oocyst contaminated soil, salads, vegetables, eating raw or undercooked meat containing tissue cysts (Sheeps, pigs,….) are the most common meat sources.

- Primary maternal infection in pregnancy–
  - ***Infant infection rate higher in 3rd trimester.***
  - ***Fetal death higher in 1st trimester***


Oral infection
Oocyst shedding
Oral infection with tissue cysts or oocysts
Sexual cycle (feline intestine)
Congenital transmission
Oocyst
♂
Asexual cycle
♀
Fertilization
Tissue cyst
Sporulated oocyst
Sporozoite
Bradyzoite
Cyst wall
Parasitophorous vacuole
Tachyzoite cycle
Tachyzoite
Infected tissues (e.g., muscle)
Chronic infection
Acute infection
Asexual cycle (mammals/birds)

Signs and sympoms

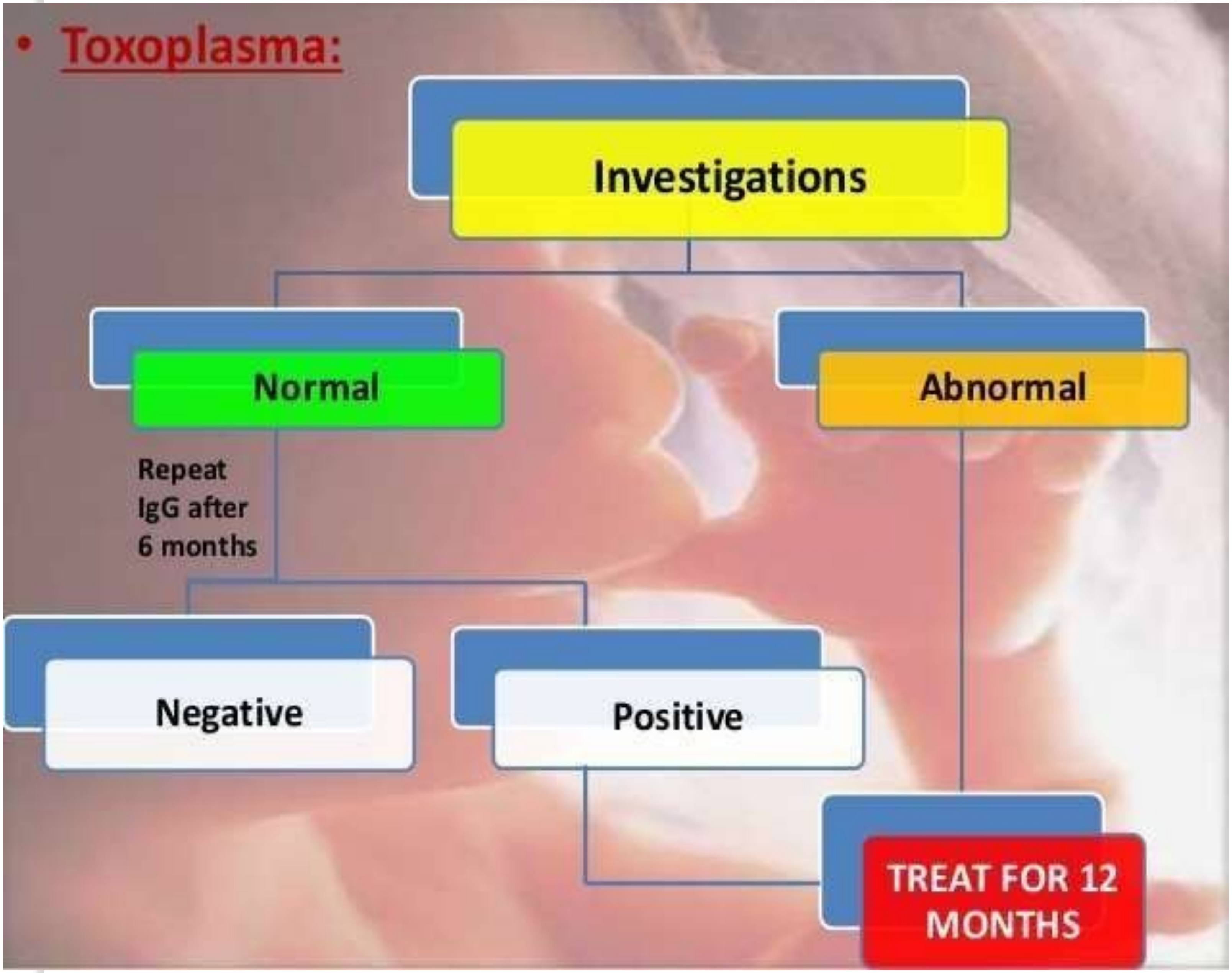
- Infected Pregnant women: 90% asymptomatic
- Incubation period is 5–18 days following exposure.
- Some may have flu- like illness, with lymphadenopathy and fatigue
- The symptoms are usually self-limiting
- Immunocompromised pregnant women are at higher risk of developing serious complications like severe encephalitis, myocarditis, pneumonitis or hepatitis.

- Similarly, infected neonates are usually asymptomatic at birth.

- The classic tetrad of findings :
  - Chorioretinitis
  - Hydrocephalus
  - Intracranial calcifications
  - Convulsion

Diagnosis

- Serial IgG measurement (for maternal infection
- Amniotic fluid PCR (for fetal infection)
- Serologic testing, brain imaging, CSF analysis and ophthalmologic evaluation (for neonatal infection)
- PCR testing of various body fluids or tissues


• Toxoplasma:
Investigations
Normal
Abnormal
Repeat IgG after 6 months
Negative
Positive
TREAT FOR 12 MONTHS

Treatment

In **PREGNANT WOMEN**

with an established recent infection, SPIRAMYCIN (3g daily in divided doses) should be given.

In **neonates** :

Pyrimethamine: 50mg twice daily for 2 days then 50mg daily. PLUS Sulfadiazine: 75mg/kg/daily in two divided doses for 2 days then 50mg/kg/twice daily PLUS Folinic Acid: 10-20mg daily

Prevention- counselling

- Avoid raw/undercooked meat
- Wash hands after gardening
- Wash raw vegetables
- Minimise contact with young kittens and their litter

RUBELLA:

- It is caused by rubella virus,Rubivirus genus and familyTogaviridae.
- Intrauterine infection with rubella virus is referred to as congenital rubella infection (CRI) or syndrome.
- **Infection with rubella earlier in pregnancy (1st trimester) cause worse prognosis and neonatal complications**.
- The virus can be transmitted to the fetus through the placenta and is capable of causing serious congenital defects, abortions, and stillbirth

In the Mother

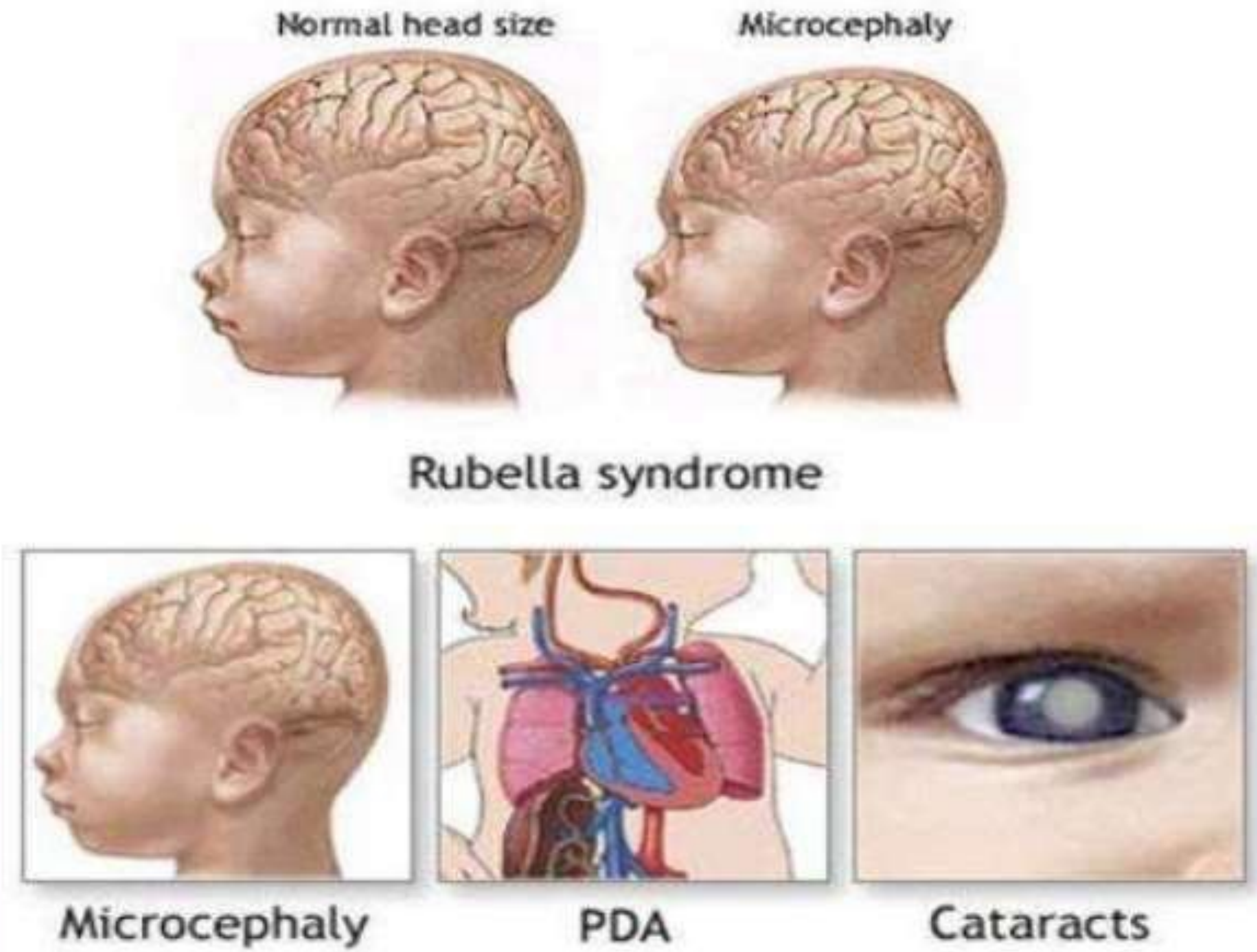
- Rubella is a mild disease
- There may be a mild prodromal illness involving a low-grade fever, malaise, coryza and mild conjunctivitis
- Lymphadenopathy may precede the rash and usually involves post-auricular and sub- occipital glands
- The rash is usually transitory, erythema tous and mostly located behind the ears and on the face and neck. Clinical diagnosis is unreliable

- Transmission to the fetus occurs via maternal hematogenous spread to the placenta.
- It typically occurs 5-7 days after maternal inoculation.
- After the virus invades the placental barrier, it spreads throughout the fetus via their vascular system.
- The congenital defects that result from infection is secondary to the cytopathical damage ensued to the blood vessels.
- This in turn results in ischemia of the affected organs

Fetal Manifestations:

Congenital rubella syndrome (CRS) presents with one or more of the following:

- Cataracts and other eye defects
- Deafness
- Cardiac abnormalities
- Microcephaly
- Retardation of intrauterine growth
- Inflammatory lesions of brain, liver, lungs and bone marrow.


Normal head size
Microcephaly
Rubella syndrome
Microcephaly
PDA
Cataracts

- **The risk of maternal-fetal transmission is the greatest in the first 10 days after gestation**
- cardiac and eye defects typically resulting when maternal infection occurs prior to 8 week.
- Hearing loss is typically observed in infections up to 18 weeks of gestation

Treatment:

- Supportive care and surveillance is theonly recommended option available at this time.
- Close monitoring within the first 6 to 12 months of life is recommended; particularly for the evaluation of hearing impairment .

Prevention:

Preventive measures include recommended immunizations, testing of pregnant women for rubella immunity and proper counseling regarding avoiding exposure

CYTOMEGALOVIRUS

- CMV is a doubles stranded DNA herpes virus
- The most common congenital viral infection.
- The CMV seropositivity rate increases with age.
- Geographic location, socioeconomic class, and work exposure are other factors that influence the risk of infection.
- CMV infection requires intimate contact through saliva, urine, and/ or other body fluids.
- Possible routes of transmission include

  1. sexual contact,   2. organ transplantation,

  3.transplacental transmission,

  4.transmission via breastmilk

  5.blood transfusion(rare)

- Primary,reactivation,or recurrent CMV infection can occur in pregnancy and can lead to congenital CMV infection.
- Approximately 85 percent of newborns with congenital CMV infection can be asymptomatic at birth.
- 15 percent will develop progressive hearing loss and visual impairment as they age.
- Transplacental infection can result in :
  - intrauterine growth restriction
  - Sensorineural hearing loss,
  - Intracranial calcifications
  - Petichiae
  - Jaudice
  - microcephaly,
  - hydrocephalus
  - hepatosplenomegaly,
  - Delayed psychomotor development,
  - Thrombocytopenia and/
  - Chorioretinitis

Vertical transmission of CMV can occur at any stage of pregnancy.

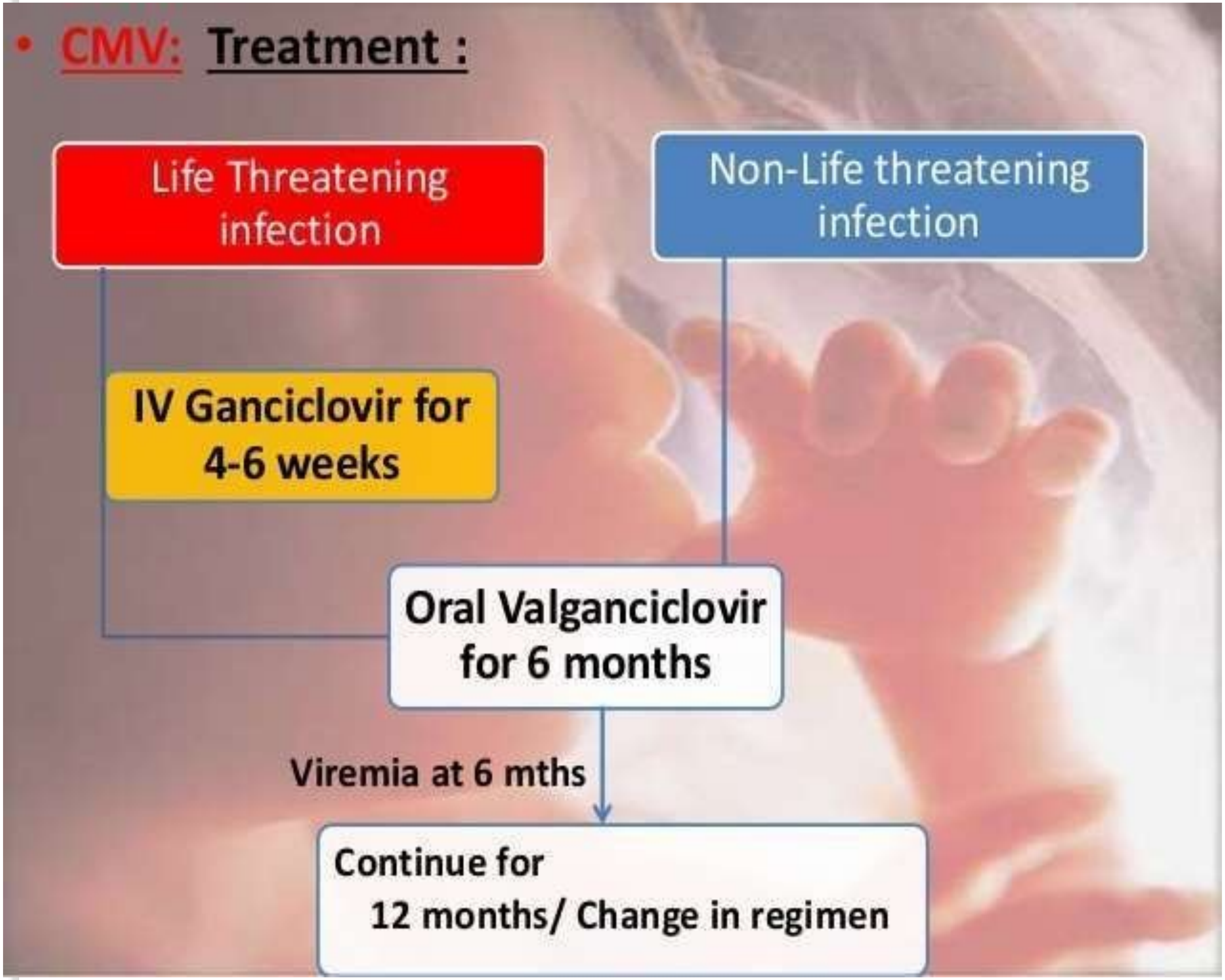
- ***Severe sequelae*** ***are more common with infection in the*** ***1st trimester***.
- ***The overall*** ***risk of infection*** ***is greatest in the*** ***3rd trimester***.

The risk of transmission to the fetus in primary infection is 30%-40%

Diagnosis :

- Virus culture from urine/saliva
- CMV-DNA PCR in urine, blood, saliva and CSF
- CMV IgM antibodies in blood before 3 weeks of age.
- IgG Avidity testing

Source: Wolff K, Goldsmith LA, Katz SI, Gilchrest BA, Paller AS, Leffell DJ: *Fitzpatrick's Dermatology in General Medicine*, 7th Edition: http://www.accessmedicine.com


CMV: Treatment :
Life Threatening infection
Non-Life threatening infection
IV Ganciclovir for 4-6 weeks
Oral Valganciclovir for 6 months
Viremia at 6 mths
Continue for 12 months/ Change in regimen

HERPES SIMPLEX VIRUS

- HSV is a member of the Herpes viridae family of viruses
- Enters the host through the inoculation of oral, genital, or conjunctival mucosa.
- Inoculation also can occur through breaks in the skin.
- Dissemination of the virus eventually allows the virus to reach the dorsal root ganglia, where it remains dormant for the rest of the host's life.
- Antiviral drugs do not affect latent HSV infection and therefore infection is life-long
- Intrauterine HSV is a rare occurrence and most likely is caused by maternal viremia associated with primary infection during pregnancy.

- Herpes simplex virus (HSV) infection during pregnancy can pose a serious threat to the developing fetus and the newborn infant.
- Transmission typically occurs via direct contact between the neonate and an infected maternal genital tract.
- If the primary HSV infection was acquired during pregnancy, then the risk of transmission is greater as compared with reactivation of a previous infection.
- Incidence of neonatal HSV infection ranges from 1 in 3200 to 1 in 10,000 births .

Intrauterine infection is associated with

- hydropsfetalis and
- in-utero fetal demise.

The characteristic ***triad*** noted at birth includes

- skin lesions consistent of vesicles,
- ulcerations or scarring ,
- eye damage and
- CNS abnormalities, such as hydranencephaly and microcephaly.

Clinical manifestation can arise any time during the first six weeks of life, but usually occurs within the first month of life

A

B

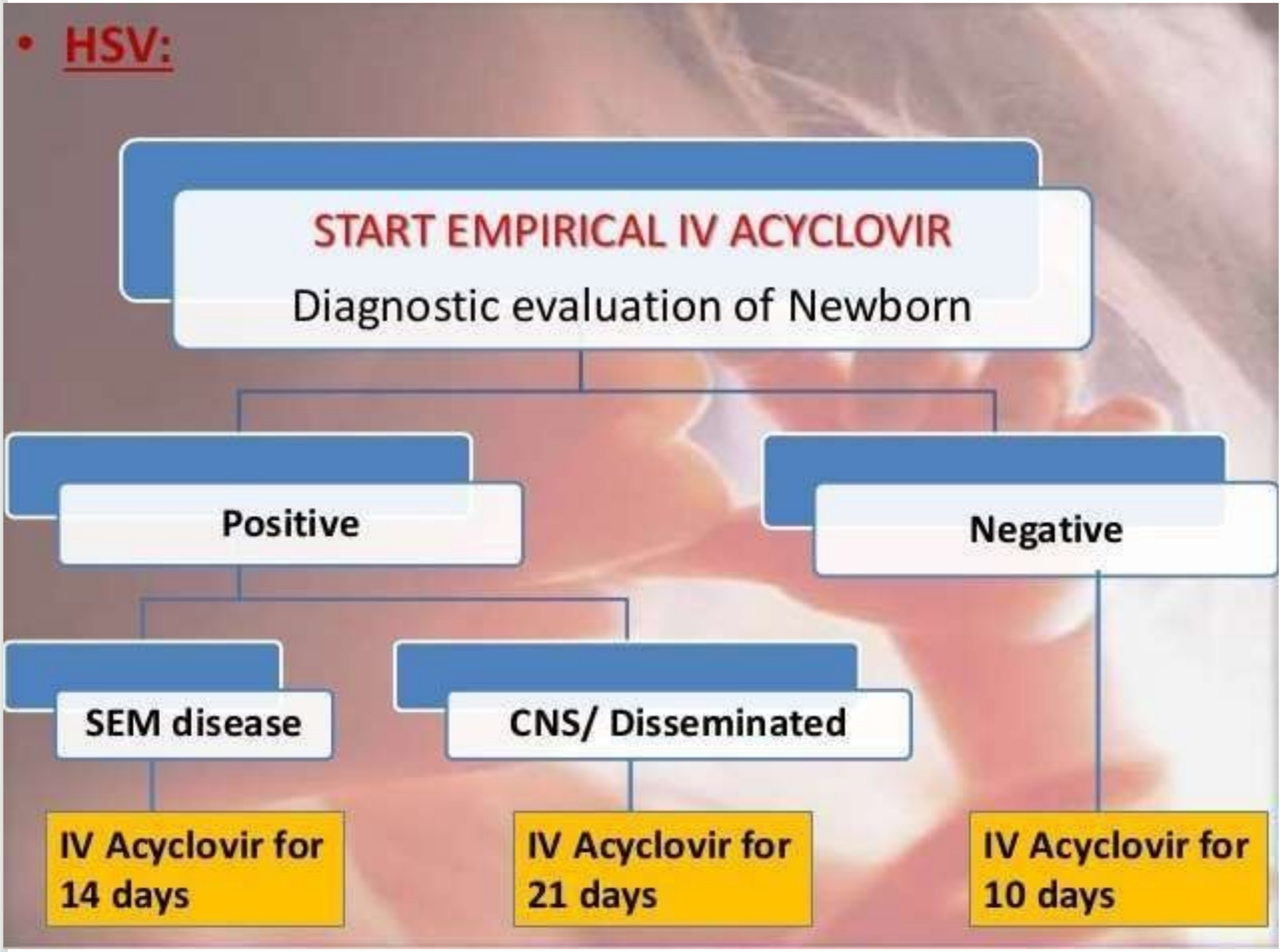
Diagnosis

- Isolation of HSV in culture
- Detection of DNA via PCR assays
- Detection of HSV specific antigens using rapid direct immunofluorescence or enzyme immunoassays.
- Classic CSF findings include :

  a mononuclear cell pleocytosis,

  normal or slightly low glucose concentration

  and moderately elevated protein level.
- Electroencephalogram (EEG) is often abnormal from early on in the disease and may show focal or multifocal periodic epileptic form discharges .
- Neuroimaging studies may show parenchymal brain edema, hemorrhage or destructive lesions in the temporal frontal, parietal or brainstem regions in the brain


• HSV:
START EMPIRICAL IV ACYCLOVIR
Diagnostic evaluation of Newborn
Positive
Negative
SEM disease
CNS/ Disseminated
IV Acyclovir for 14 days
IV Acyclovir for 21 days
IV Acyclovir for 10 days

- After completion of parenteral therapy suppressive course of oral acyclovir for 6 months

- **PREVENTION**
- 85% neonatal HSV are acquired perinatally.
- True intrauterine infection 5%
- Careful speculum examination for active genital HSV
- Caesarean section reduces risk of HSV transmission

SYPHILIS:

- Infant usually infected in utero by transplacental passage of *Treponema Pallidum* from infected mother at any time.
- Infection may also occur from contact with an infectious lesion during passage through the birth canal

- It remains unclear what factors determine which mothers, particularly those in the latent stage, will pass the disease to the fetuses. Also unclear why some infants, infected in utero, are born asymptomatic, but develop overt disease in first few wks./mo.

Infection can be transmitted to fetus at any stage of disease.

- Rate of infection **60% - 100% during second stage.**
- **Transmission rates slowly decreases with increasing duration of the disease.**

Women, untreated early syphilis: 40% of pregnancies result in spontaneous abortion, stillbirth, or perinatal death.

Clinical Manifestations

Damage to fetus depends on the stage of development at which infection has taken place and time elapsed before treatment.

- Early infection, untreated: miscarriage, stillbirth, neonatal death, IUGR, premature delivery.

- Survivors :
  - Early congenital syphilis : clinical manifestations within first 2 years of life
  - Late congenital syphilis : clinical manifestations after 2yo

EARLY CONGENITAL SYPHILIS

Early manifestations are varied, with multi-system Involvement

- Hepatosplenomegaly- diffuse inflammation, scarring
- Jaundice – due to hepatitis
- Generalized lymphadenopathy – epitrochclear nodes
- Coombs – hemolytic anemia, thromobocytopenia, leukopenia, leukocytosis
- Hydrops fetalis
- Mucocutaneous: rhinitis (highly infectious) , “snuffles”, mucous patches
- Macuolpapular rash
- Desquamation
- Pemphigus syphiliticus (vesicular bullous eruptions of palms and soles)
- Petechial lesions
- Bony lesions, osteochondritis, periostitis, pseudoparalysis
- Syphilitc leptomeningitis
- Chorioretinitis, salt and pepper fundus, glaucoma
- Pancreatitis

Congenital Syphilis

LATE CONGENITAL SYPHILIS

Results primarily from chronic inflammation of bone, teeth, and CNS.

- Interstitial keratitis (inflammatory)
- Nerve deafness
- Clutton's Joints (synositis, restricted movement)
- Hutchinson's triad (teeth, intersitial keratitis, 8 th nerve deafness)
- Mulberry molars
- Flaring scapulas
- Hydrocephalus
- Mental retardation
- Frontal bossing
- Saddle nose
- Protruding mandible ,High arched palate ,Perioral fissures
- Saber shins Anterior bowing of tibias
- Rhagades (linear scars that become fissured or ulcerated)
- Hutchinson's teeth – peg shaped upper incisors
- Frontal Bossing
- Gumma: Thin atrophic scar

Diagnosis:

- Adequacy of maternal treatment
- Examination of placenta/umbilical cord for pathology
- Dark field microscopy of suspicious lesions/body fluid
- Clinical findings suggestive of syphilis: Non immune hydrops/ jaundice/hepatosplenomegaly/ rhinitis/ skin rash
- Quantitative

Nontreponemal test: VDRL, RPR

Quantitative results correlate with disease activity, therefore helpful in screening.

Titers rise when disease is active, fall when treatment is adequate

These tests become non- reactive within a few months of adequate treatment.

Treponemal tests: TPI, FTA-ABS, MHA-TP

Treatment

- Aqueous crystalline Penicillin G 100,000-150,000U/kg/day (given q8-q12hrs) IV for 10 days

  OR

- Procaine Penicillin G 50,000 U/kg/day IM for 10days

- If >1 day of therapy missed, entire course should be restarted

VARICELLA

- Neonatal Varicella Infants whose mothers demonstrate varicella in the period from 5 days prior to delivery to 2 days afterward are at high risk for severe varicella.
- The infant acquires the infection transplacentally
- The infant's rash usually occurs toward the end of the 1st week to the early part of the 2nd week of life
- Maternal immunoglobulin G (IgG) is able to cross the placenta if delivery occurs after 30 wk of gestation

Clinical manifestations

1.Congenital varicella syndrome.

Include cicatricial skin lesions, ocular defects,limb abnormalities, CNS abnormalities, IUGR, and fetal demise or early death.

The syndrome most commonly occurs with maternal VZV infection between weeks 7 and 20 of gestation

2. Zoster.

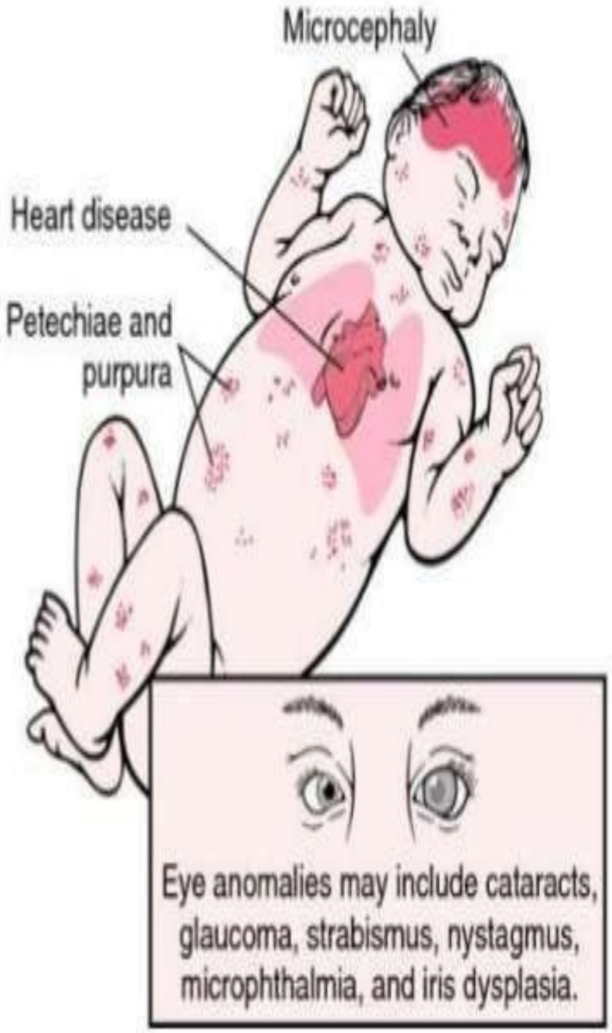
Zoster is uncommon in young infants but may occur as a consequence of *in utero fetal infection with* VZV.

Usually self-limiting, with only symptomatic therapy indicated in otherwise healthy children.

3.Postnatal varicella.

Mild disease likely due to the presence of maternal antibodies against the virus.

Rarely, severe disseminated disease occurs in newborns exposed shortly after birth following an acute maternal infection.



Herpes zoster involving the lumbar dermatome.

Period of gestation of infected mother	Outcome in the fetus
7–28 weeks	Fetal varicella syndrome
1–28 weeks	Neonatal/childhood herpes zoster
2 weeks before delivery	Neonatal chickenpox
5 days before or after delivery	Neonatal disseminated chickenpox with septicemia and increased mortality

DIAGNOSIS

- Clinical findings and maternal history
- Culture of vesicular fluid,
- VZV antibody titer by the fluorescent antibody to membrane antigen assay or by ELISA;
- Antigen detected from cells at the base of a vesicle By immunofluorescent antibody or PCR detection.

TREATMENT

-7	-6	-5	-4	-3	-2	-1	+1	+2	+3	+4

- Newborn will have protective antibodies
- Likelihood of severe disease is low
- Do not separate baby from mother
- Continue breast feeding
- No VZIG
- Acyclovir if baby develops rash

- Newborn will not have protective antibodies
- Likelihood of severe disease is high
- Separate baby from mother
- If baby devps rash stay with mother
- VZIG within 72 hours
- Acyclovir

- Newborn will not have protective antibodies
- But, likelihood of severe disease is low
- Separate baby from mother
- f baby devps rash □ stay with mother
- No VZIG
- Acyclovir if baby develops ras